

Negating quantified statements:-

eg:- $\sim (\forall x \in \mathbb{N}, P(x))$

$P(x)$ is
a function

Read:- It's not the case that $P(x)$
is true for every natural number.

$$\sim (\forall x \in \mathbb{N}, P(x))$$

$$\exists x \in \mathbb{N}, \sim P(x)$$

S: For every real number x , there is
ex:- a real number y for which $y^3 = x$.

$$S: \forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y^3 = x, Q(x)$$

$$\sim(S) = ? \quad \exists x \in \mathbb{R}, \forall y \in \mathbb{R}, y^3 \neq x \quad \checkmark$$

$$\exists x \in \mathbb{R}, \exists y \in \mathbb{R}, y^3 \neq x \quad ?$$

$$\exists x \in \mathbb{R}, \sim Q(x)$$

$$Q(x) = \exists y \in \mathbb{R}, y^3 = x$$

$$\sim(Q(x)) = \forall y \in \mathbb{R}, y^3 \neq x$$

$\sim S$: there's a real number x for which
 $y^3 \neq x$ for real number y .